

Empirical rule



1

a 68%

b 95%

c 99.7%

2

a 34%

b 47.5%

3

a 81.5%

b 97.5%

c 13.5%

d 16%

e 81.5%

f 97.5%

4

a 81.5%

b 83.85%

c 97.35%

5

a 68%

b 95%

c 99.7%

d 34%

e 13.5%

6

a 97.35%

b 449

7

a 68%

b 95%

c 99.7%

d 34%

e 13.5%

8

a 48

b 135

c $\frac{173}{3}$

d $\frac{501}{5}$

e 164

f 19

9

a 35

b 83

c 99

d 19

e 59

f $\frac{121}{3}$

10

a 340

b 499

11 95%



12

a 97.35%

b 81.5%

c 50 924

13

a 1

b 24 biscuits to 33 biscuits

14 $8.3 \text{ s} - 17.8 \text{ s}$

15 3360

16 699

17

a 74%

b 49%

z-scores

18 Mathematics

19 English

20 -2

21 46

22

a -3.7

b -3.2

c Maria

23 8.5

24 4

25

a 0.13

b 147

26

a $x = 65\%$ b $x = 66\%$ 

c 85.8%

d 72.7%

27

a 2.7

b 3.7

c Gwen

28

a -1.2 b -2.9

c Mathematics

29

a 1.2

b 3.5

c Geography

30 3

31 2

32

a 31.50

b 1.28

c 1.95

d 1.95

33

a

Mass (kg)	1	1.03	1.06	1.09	1.12
z-score	-2	-1	0	1	2

b 50%

34

a -3.3 b -4.8

c Full marathon

35

a 5.108, 5.012

b 0.3

c There is a good chance that the machine is faulty.

36

a 2.4

b 1.4

c John

37

a 27.40

e 1.63

b 1.34

f 1.8



c 2.2

g Yellowfin Bream

d 34.33